

Started on Wednesday, 25 September 2024, 12:00 PM**State** Finished**Completed on** Wednesday, 25 September 2024, 12:38 PM**Time taken** 38 mins 4 secs**Grade** 75.42 out of 100.00**Question 1**

Complete

Not graded

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- ☒ a. I confirm
- ☐ b. I confirm

The correct answers are: I confirm, I confirm

Question 2

Correct

Mark 2.50 out of 2.50

Assume you have the following categorical data in exactly this fixed order:

bird, dog, cat, fish

You are asked to create a one-hot encoding. Which of the following is the correct one-hot encoded representation of **bird**?

- ☐ a. (1, 0, 1, 1)
- ☐ b. (0, 1, 0, 0)
- ☐ c. (0, 0, 0, 1)
- ☒ d. (1, 0, 0, 0) ✓
- ☐ e. (0, 0, 0, 0)
- ☐ f. (0, 0, 1, 0)
- ☐ g. (1, 1, 1, 1)

The correct answer is: (1, 0, 0, 0)

Question 3

Correct

Mark 2.50 out of 2.50

Assume you apply the following PyTorch image transformation `transform_chain` to some example RGB image:

```
import torch
from torchvision.transforms import v2

transform_chain = v2.Compose([
    v2.PILToTensor(),
    v2.ToDtype(torch.float32, scale=True),
    v2.Resize(size=100),
    v2.ColorJitter(),
    v2.RandomRotation(degrees=180),
    v2.RandomVerticalFlip(p=0.5),
    v2.RandomHorizontalFlip(p=0.5)
])
```

Which of the following statements are correct?

- ☒ a. The resulting image might be vertically flipped. ✓
- ☒ b. The resulting image might be horizontally flipped. ✓
- ☒ c. The resulting image might have its colors changed. ✓
- ☒ d. The resulting image might be rotated. ✓

The correct answers are: The resulting image might have its colors changed., The resulting image might be horizontally flipped., The resulting image might be vertically flipped., The resulting image might be rotated.

Question 4

Correct

Mark 2.50 out of 2.50

The `forward` method of a class derived from `torch.nn.Module` ...

- ☐ a. ... computes the gradients.
- ☐ b. ... sets up the module architecture.
- ☒ c. ... specifies how an input is transformed into an output (the "flow" through the architecture). ✓
- ☐ d. ... performs one weight update during training.
- ☐ e. ... pushes a PyTorch module to a given device.

The correct answer is: ... specifies how an input is transformed into an output (the "flow" through the architecture).

Question 5

Correct

Mark 2.50 out of 2.50

Which of the following statements are correct regarding scripting and tracing to create TorchScript programs?

- ☐ a. Scripting and tracing cannot be combined.
- ☒ b. Tracing requires an example input. ✓
- ☒ c. Both scripting and tracing turn Python code into TorchScript code. ✓
- ☐ d. With scripting, control flow such as branching is lost.

The correct answers are: Both scripting and tracing turn Python code into TorchScript code., Tracing requires an example input.

Question 6

Partially correct

Mark 1.67 out of 2.50

Consider the following `Dataset` implementation:

```
import numpy as np
from torch.utils.data import Dataset

class MyDataset(Dataset):
    def __getitem__(self, index):
        rng = np.random.default_rng(seed=index)
        return index, rng.random()
```

Which of the following statements are correct?

- ☐ a. For a given `index`, the data set always returns the same sample.
- ☒ b. An instance of this class would produce infinitely many samples. ✓
- ☐ c. An instance of this class cannot be processed by a `torch.utils.data.DataLoader` because of the randomness.
- ☒ d. A sample returned by the `__getitem__` method always contains the index and a randomly generated number. ✓

The correct answers are: An instance of this class would produce infinitely many samples., A sample returned by the `__getitem__` method always contains the index and a randomly generated number., For a given `index`, the data set always returns the same sample.

Question 7

Correct

Mark 2.50 out of 2.50

What is the output of the following code?

```
import torch

class MyModule(torch.nn.Module):
    def __init__(self, x):
        super().__init__()
        self.x = x

my_module = MyModule(2)
result = my_module(5)
print(result)
```

- ☐ a. 2
- ☒ b. It raises an exception because `MyModule` does not have a `forward` method. ✓
- ☐ c. 10
- ☐ d. It raises an exception because `MyModule` does not have any trainable parameters.
- ☐ e. It raises an exception because `self.x` is not a (sub)module.
- ☐ f. 7
- ☐ g. 5

The correct answer is: It raises an exception because `MyModule` does not have a `forward` method.

Question 8

Correct

Mark 2.50 out of 2.50

Which of the following are benefits of a version control system (VCS) such as git?

- ☐ a. Improved performance for deep neural networks.
- ☒ b. Useful for collaboration. ✓
- ☐ c. Makes hyperparameter tuning easier.
- ☒ d. Efficient tracking of files and their changes. ✓

The correct answers are: Efficient tracking of files and their changes., Useful for collaboration.

Question 9

Correct

Mark 2.50 out of 2.50

Assume a classification task where you want to classify handwritten digits (0-9) which are available as grayscale images. Which of the following data augmentation techniques are useful?

- ☒ a. Adding slight noise. ✓
- ☒ b. Rotating by ± 5 degrees. ✓
- ☐ c. Modifying color channels.
- ☐ d. Flipping vertically.

The correct answers are: Rotating by ± 5 degrees., Adding slight noise.

Question 10

Correct

Mark 2.50 out of 2.50

Which of the following statements are correct when talking about normalization/scaling?

- ☐ a. Applying normalization/scaling is always beneficial, regardless of the used methods/models.
- ☐ b. Normalization/scaling cannot be used in conjunction with cross validation.
- ☒ c. When applying normalization/scaling in a setting with a dedicated training and test set, care must be taken which data sets to use for computing normalization constants. ✓
- ☒ d. Normalization/scaling depends on the data set. ✓

The correct answers are: Normalization/scaling depends on the data set., When applying normalization/scaling in a setting with a dedicated training and test set, care must be taken which data sets to use for computing normalization constants.

Question 11

Incorrect

Mark 0.00 out of 2.50

Which of the following statements are true regarding data augmentation?

- ☒ a. Data augmentation reduces underfitting. ✗
- ☒ b. Data augmentation can have a negative impact on the model performance. ✓
- ☒ c. Data augmentation is about creating "new" artificial samples by modifying existing samples. ✓
- ☒ d. Data augmentation can have a positive impact on the model performance. ✓

The correct answers are: Data augmentation is about creating "new" artificial samples by modifying existing samples., Data augmentation can have a negative impact on the model performance., Data augmentation can have a positive impact on the model performance.

Question 12

Correct

Mark 2.50 out of 2.50

The special method `__getitem__` in a `torch.utils.data.Dataset` derived class should ...

- ☐ a. ... return all test samples.
- ☐ b. ... return minibatched samples.
- ☒ c. ... return one sample. ✓
- ☐ d. ... return all training samples.

The correct answer is: ... return one sample.

Question 13

Incorrect

Mark 0.00 out of 2.50

Training a neural network for one update will ...

- ☐ a. ... perform training until early stopping is applied.
- ☐ b. ... perform one weight update.
- ☒ c. ... perform one training iteration over all training samples. ✗
- ☐ d. ... perform training for a fixed number of iterations.

The correct answer is: ... perform one weight update.

Question 14

Incorrect

Mark 0.00 out of 2.50

You are counting the number of completed home exercises. Which type of data is this?

- ☐ a. Discrete numerical data
- ☒ b. Continuous numerical data ✗
- ☐ c. Categorical data
- ☐ d. Ordinal data

The correct answer is: Discrete numerical data

Question 15

Correct

Mark 2.50 out of 2.50

Assume you want to normalize your data and compute a normalization constant from your data set (e.g., the mean over many samples). What would be the correct way to proceed in terms of training, validation and test set?

- ☒ a. Determine the constant for normalization on the training data and apply it also to the validation and test data. ✓
- ☐ b. Determine the constant for normalization on the validation and test data and apply it also to the training data.
- ☐ c. Determine the constant for normalization on the whole data set and apply it to training, validation and test data.
- ☐ d. Determine the constant for normalization on the training and test data and apply it also to the validation data.
- ☐ e. Determine the constant for normalization on the training and validation data and apply it also to the test data.

The correct answer is: Determine the constant for normalization on the training data and apply it also to the validation and test data.

Question 16

Correct

Mark 2.50 out of 2.50

The `__init__` method of a class derived from `torch.nn.Module` ...

- ☒ a. ... sets up the module architecture. ✓
- ☐ b. ... specifies how an input is transformed into an output (the "flow" through the architecture).
- ☐ c. ... initializes all submodules based on the superclass.
- ☐ d. ... initializes the training procedure.

The correct answer is: ... sets up the module architecture.

Question 17

Correct

Mark 2.50 out of 2.50

What is the shape of the output tensor `result` when running the following code?

```
import torch

class MyModule(torch.nn.Module):
    def __init__(self):
        super().__init__()
        self.linear = torch.nn.Linear(in_features=64, out_features=32)

    def forward(self, x):
        out = self.linear(x)
        return torch.sigmoid(out)

inp = torch.rand(size=(128, 64))
my_module = MyModule()
result = my_module(inp)
```

- ☐ a. (64,)
- ☐ b. (128, 64, 32)
- ☐ c. (128, 64)
- ☐ d. (64, 32)
- ☒ e. (128, 32) ✓
- ☐ f. (128,)
- ☐ g. (32,)

The correct answer is:

(128, 32)

Question 18

Correct

Mark 2.50 out of 2.50

Which of the following statements are correct regarding TorchScript?

- ☒ a. TorchScript code can provide a performance speed-up. ✓
- ☒ b. Every TorchScript code is also valid Python code. ✓
- ☐ c. Every Python code is also valid TorchScript code.
- ☐ d. TorchScript eases the implementation of neural networks compared to plain Python.

The correct answers are: Every TorchScript code is also valid Python code., TorchScript code can provide a performance speed-up.

Question 19

Correct

Mark 2.50 out of 2.50

In a supervised setting, how can one determine the prediction performance of some machine learning model?

- ☐ a. By running a performance profiler on the model.
- ☐ b. By averaging the gradients of the model and comparing it to a specified threshold.
- ☒ c. By using a loss function to compute the deviation between the model prediction and the true target. ✓
- ☐ d. By comparing the model input and the model prediction with each other.

The correct answer is: By using a loss function to compute the deviation between the model prediction and the true target.

Question 20

Correct

Mark 2.50 out of 2.50

Which of the following are typical data normalization/scaling approaches?

- ☒ a. Scaling to zero (0) mean and unit (1) variance. ✓
- ☒ b. Scaling to range [-1, 1]. ✓
- ☒ c. Scaling to range [0, 1]. ✓
- ☐ d. Scaling to range [-infinity, +infinity].

The correct answers are: Scaling to range [0, 1]., Scaling to range [-1, 1]., Scaling to zero (0) mean and unit (1) variance.

Question 21

Correct

Mark 2.50 out of 2.50

Which of the following are common steps when designing a machine learning project?

- ☒ a. Assessing which software and hardware can and/or should be available. ✓
- ☒ b. Checking related work to see if similar problems have already been studied. ✓
- ☒ c. Getting a clear definition and understanding of the project goal. ✓
- ☐ d. Starting a hyperparameter search for a neural network model because this might take a long time and is thus recommended to start early, i.e., in the design phase.

The correct answers are: Getting a clear definition and understanding of the project goal., Assessing which software and hardware can and/or should be available., Checking related work to see if similar problems have already been studied.

Question 22

Correct

Mark 2.50 out of 2.50

Which of the following is the correct order of performing one weight update in PyTorch (you can assume that `optimizer` is a valid PyTorch optimizer object)?

- ☐ a. 1. Compute `loss`
2. Update weights `optimizer.step()`
3. Compute gradients `loss.backward()`
4. Reset gradients `optimizer.zero_grad()`
- ☐ b. 1. Compute `loss`
2. Compute gradients `loss.backward()`
3. Reset gradients `optimizer.zero_grad()`
4. Update weights `optimizer.step()`
- ☐ c. 1. Reset gradients `optimizer.zero_grad()`
2. Compute `loss`
3. Update weights `optimizer.step()`
4. Compute gradients `loss.backward()`
- ☒ d. 1. Compute `loss` ✓
2. Compute gradients `loss.backward()`
3. Update weights `optimizer.step()`
4. Reset gradients `optimizer.zero_grad()`

The correct answer is: 1. Compute `loss`
2. Compute gradients `loss.backward()`
3. Update weights `optimizer.step()`
4. Reset gradients `optimizer.zero_grad()`

Question 23

Correct

Mark 2.50 out of 2.50

When is a model considered to be overfitting?

- ☐ a. A model is overfitting if its training loss becomes increasingly large.
- ☒ b. A model is overfitting if it fits the training data too well, which leads to decreased generalization performance. ✓
- ☐ c. A model is overfitting if the number of parameters approaches the number of hyperparameters.
- ☐ d. A model is overfitting if it does not fit the training data well at all, which leads to decreased generalization performance.

The correct answer is: A model is overfitting if it fits the training data too well, which leads to decreased generalization performance.

Question 24

Partially correct

Mark 1.67 out of 2.50

Which of the following statements regarding a `torch.utils.data.DataLoader` are correct?

- ☒ a. It extracts minibatches of samples from a `torch.utils.data.Dataset` instance. ✓
- ☒ b. It supports shuffling. ✓
- ☐ c. It supports multi-processing.
- ☐ d. It splits the data into training and test sets.

The correct answers are: It extracts minibatches of samples from a `torch.utils.data.Dataset` instance., It supports shuffling., It supports multi-processing.

Question 25

Partially correct

Mark 0.83 out of 2.50

Which of the following statements are correct regarding minibatch learning?

- ☐ a. Minibatch tries to combine the benefits of online learning and full-batch learning.
- ☐ b. A single training sample is used for one update.
- ☒ c. `b` samples are used for one update, where `b` is the batch size. ✓
- ☐ d. The batch size is a hyperparameter.

The correct answers are: The batch size is a hyperparameter., `b` samples are used for one update, where `b` is the batch size., Minibatch tries to combine the benefits of online learning and full-batch learning.

Question 26

Incorrect

Mark 0.00 out of 2.50

Which statements regarding PyTorch's `torch.nn.Module` are correct?

- ☒ a. Many predefined PyTorch modules exist. ✓
- ☒ b. It is the base class for all data sets. ✗
- ☒ c. It can contain and utilize other modules. ✓
- ☒ d. It can contain trainable parameters (`torch.nn.Parameter`). ✓

The correct answers are: It can contain and utilize other modules., Many predefined PyTorch modules exist., It can contain trainable parameters (`torch.nn.Parameter`).

Question 27

Incorrect

Mark 0.00 out of 2.50

Which of the following statements are correct regarding the following function and with respect to TorchScript?

```
def function(x):  
    if x.min() <= 0:  
        return 0  
    return x
```

- ☒ a. Scripting the function is possible. ✖
- ☐ b. Tracing the function is not possible.
- ☒ c. Tracing the function is possible. ✔
- ☐ d. Scripting the function is not possible.

The correct answers are: Tracing the function is possible., Scripting the function is not possible.

Question 28

Incorrect

Mark 0.00 out of 2.50

What is a hyperparameter in the setting of machine learning?

- ☐ a. It represents the process of training and evaluating an algorithm.
- ☐ b. It represents a user-specified parameter that typically influences the model and/or training procedure.
- ☐ c. It represents the number of model parameters.
- ☒ d. It represents an internal model parameter that is adjusted during training. ✖

The correct answer is: It represents a user-specified parameter that typically influences the model and/or training procedure.

Question 29

Correct

Mark 2.50 out of 2.50

Which of the following statements must be taken into consideration when doing a literature review regarding a new machine learning project?

- ☒ a. Wikis, blog posts, reddit and other similar sites can provide a good overview and introduction ✔
- ☒ b. Peer-reviewed literature can be a good way to get state-of-the-art information. ✔
- ☒ c. There can be research biases. ✔
- ☐ d. If there is no matching literature, it means that the project cannot be done using machine learning.

The correct answers are: Peer-reviewed literature can be a good way to get state-of-the-art information., Wikis, blog posts, reddit and other similar sites can provide a good overview and introduction, There can be research biases.

Question 30

Correct

Mark 2.50 out of 2.50

Assume that `MyModule` is a class properly derived from the `torch.nn.Module` class and `x`, `y`, `z` are PyTorch tensors. Which of the following statements are correct regarding the following code?

```
my_module = MyModule(x, y)
my_module(z)
```

- ☐ a. It creates an instance of `MyModule` and calls the `__init__` method with argument `z`.
- ☒ b. It (ultimately) calls the `forward` method of `my_module` with argument `z`. ✓
- ☐ c. It (ultimately) calls the `forward` method of `my_module` with arguments `x`, `y`.
- ☐ d. It creates an instance of `MyModule` and calls the `__init__` method with arguments `x`, `y`, `z`.
- ☒ e. It creates an instance of `MyModule` and calls the `__init__` method with arguments `x`, `y`. ✓
- ☐ f. It (ultimately) calls the `forward` method of `my_module` with arguments `x`, `y`, `z`.

The correct answers are: It creates an instance of `MyModule` and calls the `__init__` method with arguments `x`, `y`., It (ultimately) calls the `forward` method of `my_module` with argument `z`.

Question 31

Partially correct

Mark 1.25 out of 2.50

You are training a neural network model using some gradient-based iterative method. The first training loss that you compute is rather high. Continuing the training process then leads to a gradually lower loss. Which of the following statements are correct?

- ☒ a. The training appears to be working, i.e., the performance is improving. ✓
- ☐ b. The gradients are zero.
- ☐ c. The model is getting continuously worse, since the loss should get higher during training.
- ☐ d. The model parameters are changing.

The correct answers are: The training appears to be working, i.e., the performance is improving., The model parameters are changing.

Question 32

Correct

Mark 2.50 out of 2.50

What is meant by the term "regularization" in the context of machine learning?

- ☐ a. Techniques to reduce underfitting.
- ☒ b. Techniques to reduce overfitting. ✓
- ☐ c. Using cross entropy as loss.
- ☐ d. Replacing non-linear activation functions with linear activation functions.

The correct answer is: Techniques to reduce overfitting.

Question 33

Correct

Mark 2.50 out of 2.50

What is the output of the following code?

```
import torch

def function(x: torch.Tensor):
    return -x if x.sum() < 0 else x

scripted_function = torch.jit.script(function)
traced_function = torch.jit.trace(function, example_inputs=torch.tensor([0, -1]))

actual_input = torch.tensor([1, 2, 3])
print("s:", scripted_function(actual_input).tolist())
print("t:", traced_function(actual_input).tolist())
```

- ☐ a.

s: [1, 2, 3]
t: [0, 1]
- ☐ b.

s: [-1, -2, -3]
t: [-1, -2, -3]
- ☐ c.

s: [-1, -2, -3]
t: [1, 2, 3]
- ☐ d.

s: [1, 2, 3]
t: [0, -1]
- ☒ e.

s: [1, 2, 3]
t: [-1, -2, -3]

 ✓
- ☐ f.

s: [-1, -2, -3]
t: [0, -1]
- ☐ g.

s: [-1, -2, -3]
t: [0, 1]
- ☐ h.

s: [1, 2, 3]
t: [1, 2, 3]

The correct answer is:

s: [1, 2, 3]
t: [-1, -2, -3]

Question 34

Incorrect

Mark 0.00 out of 2.50

Assume you have the following binary confusion matrix:

		Predicted	
		positive	negative
Actual	positive	40	0
	negative	55	5

Which of the following statements are correct?

- ☐ a. The model is more likely to predict the positive class.
- ☐ b. The model never made an incorrect classification regarding the actually negative samples.
- ☒ c. The model never made an incorrect classification regarding the actually positive samples. ✓
- ☒ d. The classification performance is close to perfect (only off by 5 samples). ✗

The correct answers are: The model never made an incorrect classification regarding the actually positive samples., The model is more likely to predict the positive class.

Question 35

Correct

Mark 2.50 out of 2.50

Which of the following problems might occur when working with a new data set?

- ☒ a. Empty or corrupted files. ✓
- ☒ b. Missing datapoints. ✓
- ☒ c. Outlier values. ✓
- ☒ d. Incorrect labeling. ✓
- ☒ e. Duplicated datapoints. ✓
- ☒ f. Wrong filetypes. ✓

The correct answers are: Empty or corrupted files., Wrong filetypes., Duplicated datapoints., Missing datapoints., Outlier values., Incorrect labeling.

Question 36

Correct

Mark 2.50 out of 2.50

What does independently and identically distributed (i.i.d.) mean?

- ☒ a. Each sample has the same probability distribution as the others and all are mutually independent. ✓
- ☐ b. Each sample has the same probability distribution as the others and all are mutually dependent.
- ☐ c. Each sample has a different probability distribution as the others and all are mutually independent.
- ☐ d. Each sample has a different probability distribution as the others and all are mutually dependent.

The correct answer is: Each sample has the same probability distribution as the others and all are mutually independent.

Question 37

Correct

Mark 2.50 out of 2.50

Which of the following statements are correct regarding a hash function?

- ☒ a. A hash function should ideally be fast to compute. ✓
- ☐ b. A hash function is used to compute a variably sized data vector for a given input.
- ☐ c. A hash function is typically used as a non-linear activation function.
- ☐ d. A hash function can only be used for files.

The correct answer is: A hash function should ideally be fast to compute.

Question 38

Incorrect

Mark 0.00 out of 2.50

Which statements regarding PyTorch's automatic parameter registration of a `torch.nn.Module` are correct?

- ☒ a. When assigned as attribute, a `torch.nn.Parameter` will automatically be registered. ✓
- ☐ b. When assigned as attribute, a `torch.nn.Module`'s trainable parameters will automatically be registered.
- ☒ c. When assigned as attribute, a `torch.Tensor` will automatically be registered. ✗
- ☒ d. When assigned as attribute, a `torch.utils.data.Dataset`'s parameters will automatically be registered. ✗

The correct answers are: When assigned as attribute, a `torch.nn.Parameter` will automatically be registered., When assigned as attribute, a `torch.nn.Module`'s trainable parameters will automatically be registered.

Question 39

Correct

Mark 2.50 out of 2.50

Which of the following statements are correct regarding numerical data?

- ☒ a. Can be discrete. ✓
- ☒ b. Has a quantative meaning. ✓
- ☐ c. Has a qualitative meaning.
- ☒ d. Can be continuous. ✓

The correct answers are: Has a quantative meaning., Can be continuous., Can be discrete.

Question 40

Correct

Mark 2.50 out of 2.50

Which of the following statements are correct regarding monitoring a model during training?

- ☒ a. It enables an easier detection of over- or underfitting. ✓
- ☐ b. It reduces the time needed for training.
- ☒ c. It allows inspecting weights and gradients and how they change during training. ✓
- ☐ d. It increases the model performance.

The correct answers are: It allows inspecting weights and gradients and how they change during training., It enables an easier detection of over- or underfitting.

Question 41

Correct

Mark 2.50 out of 2.50

What is the purpose of a loss function?

- ☒ a. To compute the difference between the model output and the actual targets (ground truth). ✓
- ☐ b. To compute the quality of the input data.
- ☐ c. To compute the gradients for gradient-based iterative methods.
- ☐ d. To compute the run-time performance of a model.

The correct answer is: To compute the difference between the model output and the actual targets (ground truth).